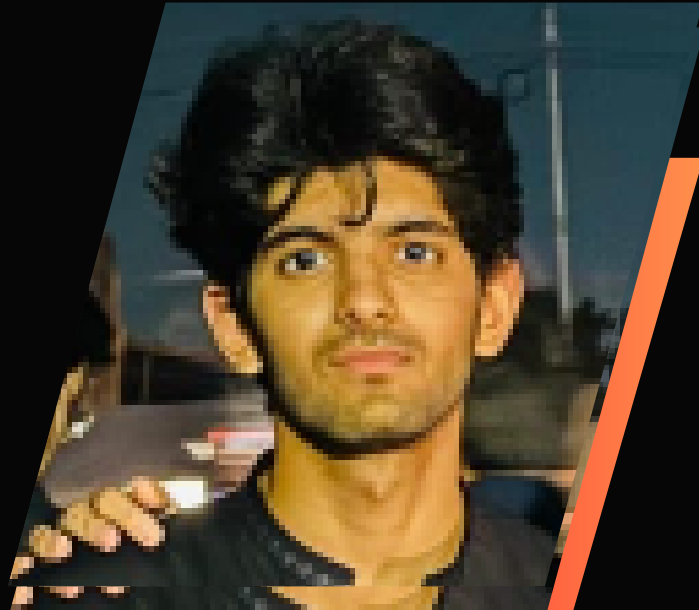


VERIDRIVE

AI-POWERED RISK INTELLIGENCE PLATFORM

<https://github.com/S-Huzaifa-Ali/Devops-Project>

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PROBLEM STATEMENT

NO BEHAVIORAL SCREENING

Traditional rental agencies rely solely on age, license, and credit score — missing critical behavioral signals that predict risk on the road.

HIGH FRAUD & DAMAGE LOSS

Rental companies lose billions annually to high-risk drivers who slip through manual screening. Fraudulent bookings and vehicle damage go undetected until it's too late.

SLOW MANUAL REVIEW

Human review teams take hours or days to process applications, creating bottlenecks, poor customer experience, and inconsistent decisions.

NO EXPLAINABILITY

Black-box rejection systems leave customers with no explanation. Regulators and applicants demand transparent, auditable risk decisions.

HOW VERIDRIVE SOLVES THE PROBLEM

01 Mastodon OAuth

Applicants authorize via Mastodon. VeriDrive fetches last 10 posts + bio without storing credentials.

02 NLP Risk Engine

TF-IDF vectorization + multi-class classifier detects Hate, Offensive, or Neither across post content.

03 Risk Score 0–100

Composite score drives auto-approval (0–30), admin review (31–65), or auto-decline (66–100).

04 SHAP Explainability

Every decision comes with SHAP feature importance so admins and applicants understand the reasoning.

05 Continuous Training

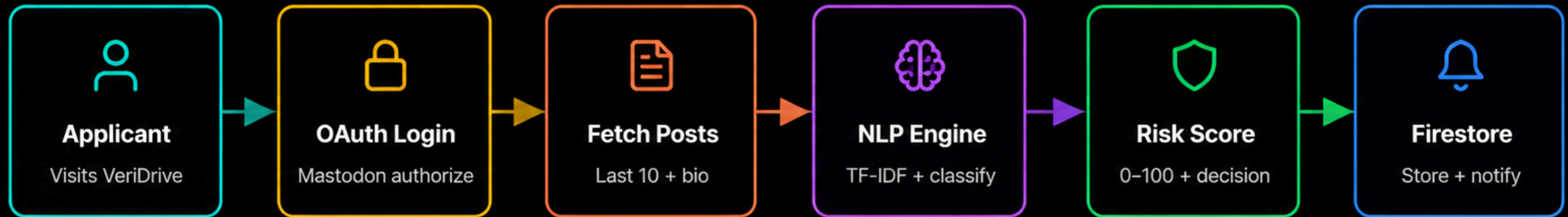
Admin overrides feed back into the model. KL-divergence drift detection triggers auto-retraining.

06 CI/CD Deployment

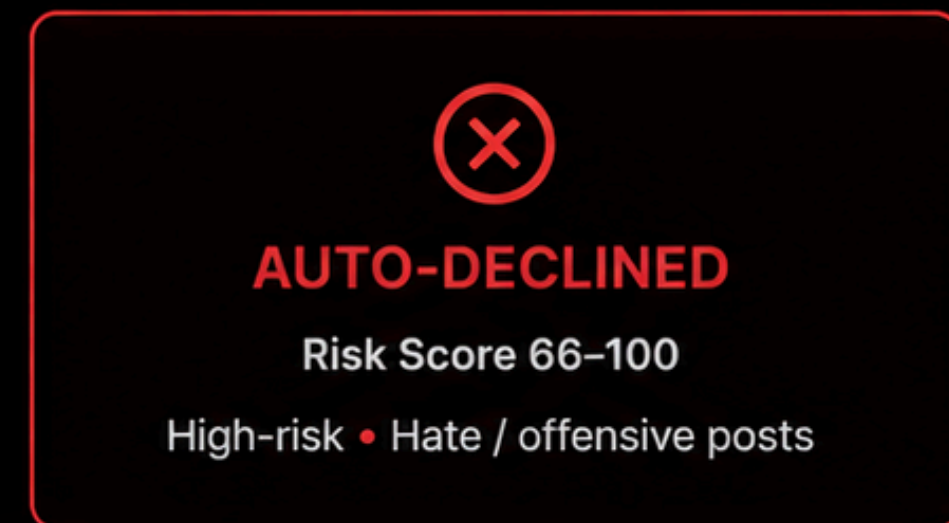
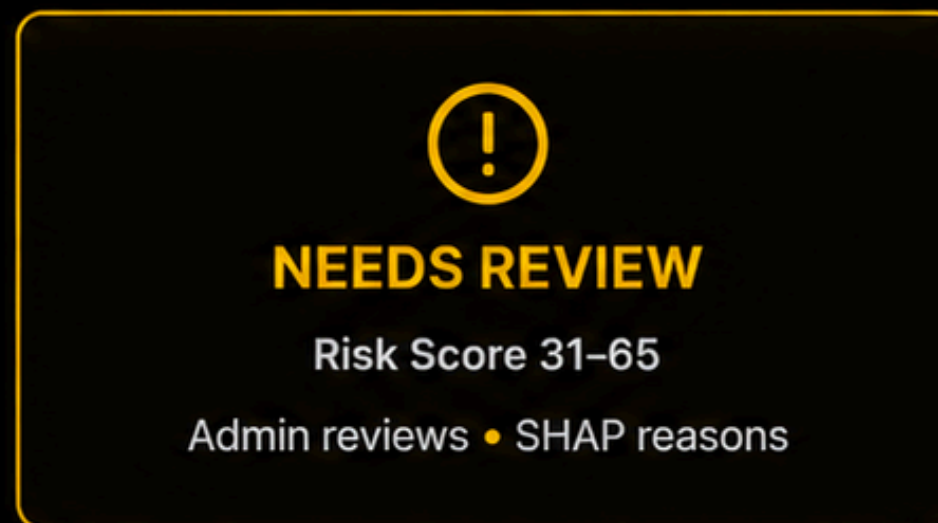
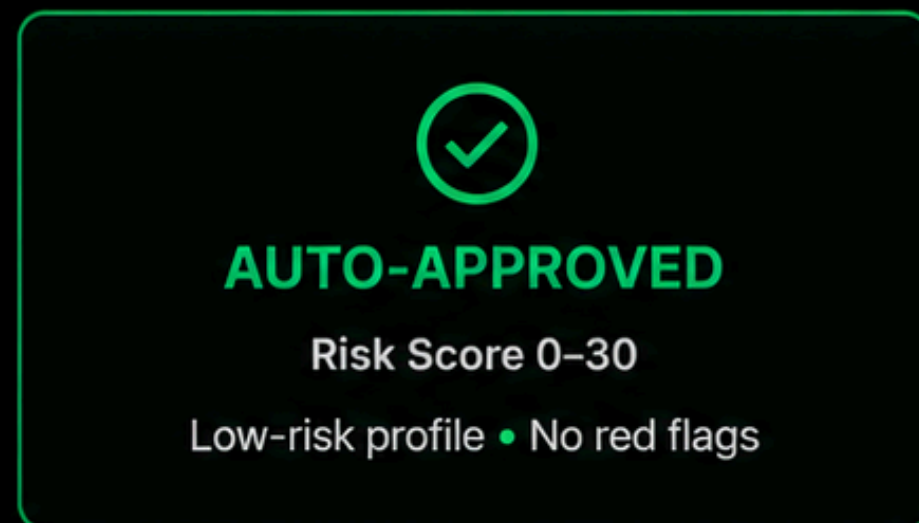
GitHub Actions pipeline: test → build → push Docker → deploy on every commit. Zero-downtime releases.

END-TO-END DATA FLOW

From Mastodon profile to risk decision in < 2 seconds



DECISION OUTCOMES



MLOPS LIFECYCLE

01

Data Ingestion

Primary Dataset

gilani-2017.tsv — labeled tweets

Live Mastodon Posts

Last 10 posts via OAuth access token

Admin Overrides

POST /admin/overrides → Firestore

Data Versioning

DVC-compatible file tracking

TOOLS & FILES

gilani-2017.tsv · Mastodon API
Firestore · pipeline_utils.py

02

Preprocessing

HTML Stripping

`re.sub(r'<[^>]*>', '', text)`

Tokenization

Split, lowercase, remove stopwords

TF-IDF Vectorization

n-gram range (1,2), max_features=5000

Feature Concatenation

Bio vector + Posts vector → feature X

TOOLS & FILES

pipeline_utils.py · scikit-learn
TfidfVectorizer · NLTK stopwords

03

Model Training

Multi-Class Training

Hate / Offensive / Neither (3 classes)

Cross-Validation

StratifiedKFold · F1-macro scoring

SHAP Explainability

TreeExplainer → shap_summary.png

Model Serialization

`pickle.dump()` → sentiment_model.pkl

TOOLS & FILES

train_sentiment.py · promote_model.py
SHAP · sklearn · sentiment_model.pkl

04

Evaluation

Confusion Matrix

confusion_matrix.png generated

Precision / Recall / F1

Per-class + macro average report

pytest Regression

test_regression.py · test_sentiment.py

Promotion Gate

$F1 \geq \text{threshold}$ → promote_model.py

TOOLS & FILES

confest.py · pytest · sklearn metrics
confusion_matrix.png · F1 threshold

05

Deployment

Docker Build

Multi-stage Dockerfile · HEALTHCHECK

GitHub Actions CI/CD

push → test → build → push → deploy

FastAPI on Uvicorn

Port 8000 · /health · CORS configured

Container Registry

`docker push latest` → production slot

TOOLS & FILES

Dockerfile · .github/workflows/ · Vercel
docker-compose.yml · uvicorn main:app

06

Monitor + CT Loop

Drift Detection

drift_monitor.py · KL-divergence check

Performance Tracking

Confidence score trends in Firestore

Auto-Retrain Trigger

ct_loop.py → collects new data → trains

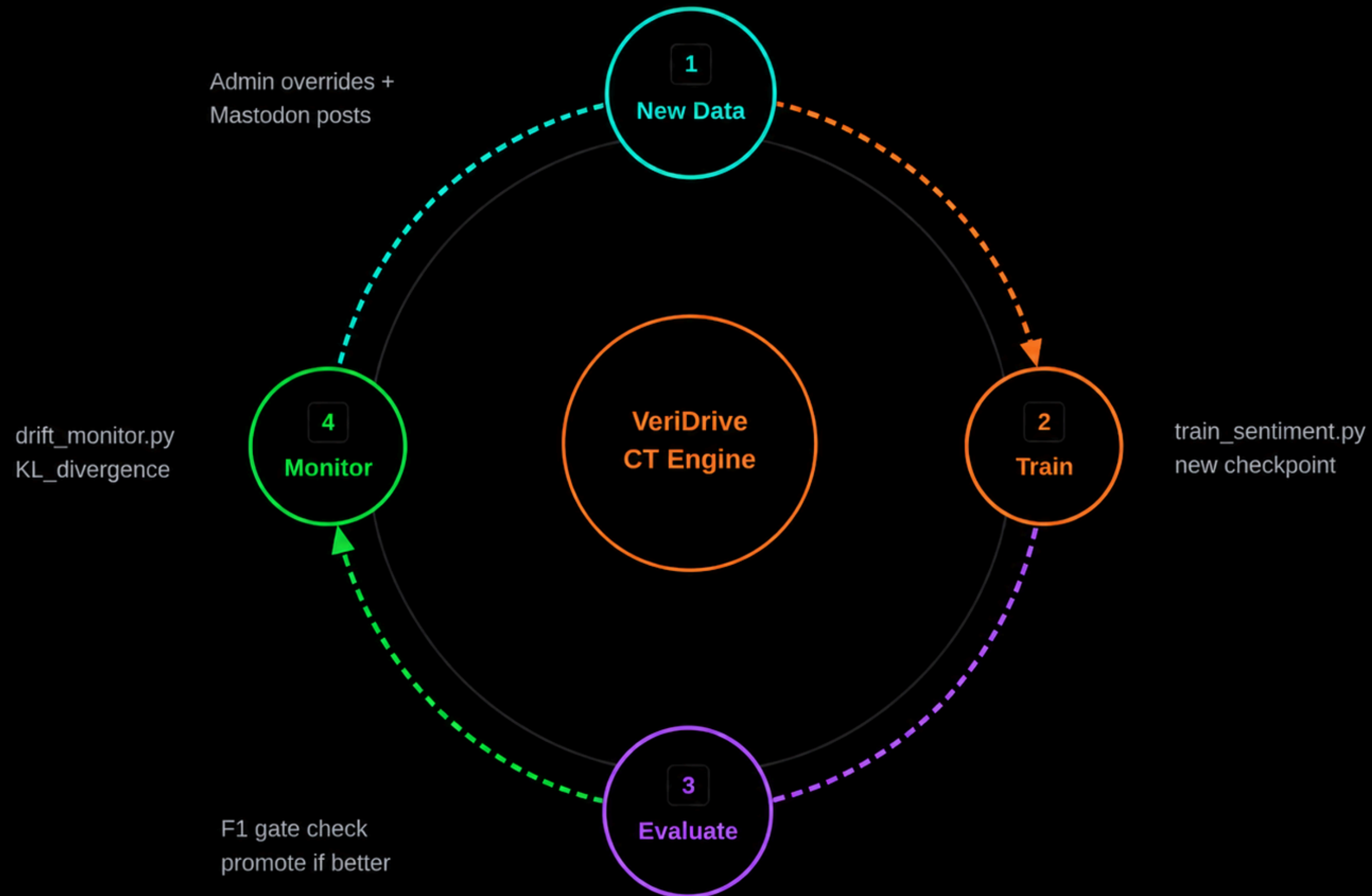
Auto-Promote Winner

promote_model.py if F1 improved

TOOLS & FILES

drift_monitor.py · ct_loop.py
promote_model.py · Firestore logs

CONTINUOUS TRAINING CYCLE



CI/CD PIPELINE

TRIGGER

GIT PUSH

- > push → main
- > pull_request
- > manual dispatch



TEST

PYTEST SUITE

- > test_sentiment.py
- > test_regression.py
- > F1 promotion gate



BUILD

DOCKER IMAGE

- > Multi-stage Dockerfile
- > HEALTHCHECK /health
- > Push → registry



DEPLOY

FASTAPI + VERCEL

- > Backend: port 8000
- > Frontend: Vercel
- > CORS configured



MONITOR

DRIFT + CT LOOP

- > KL-divergence check
- > Firestore confidence
- > Auto-retrain trigger



.github/workflows/
CI/CD action definitions



Dockerfile
Multi-stage container build



promote_model.py
F1 gate → auto-promote winner

DEMO

localhost:8081/signup

VERIDRIVE
RISK · INTELLIGENCE

OVERVIEW ADMIN HOW IT WORKS

LOG IN SIGN UP RUN CHECK

ACCOUNT

Sign Up

Create an account to submit posts for rental risk analysis.

DISPLAY NAME

Your name

EMAIL

you@example.com

PASSWORD

Min 6 characters

CREATE ACCOUNT

Already have an account? [Log in](#)

34°C Sunny

Search

3:28 PM 5/8/2026